Chapter 12: The Logistic Map

November 2020

1 Background

The chapter I was interested in for this project was chapter 12, which covers nonlinear mechanics and chaos. The problem I was assigned was the Logistic Map. This system has very simple equations of motion, compared to the mechanical systems covered in the previous sections of the textbook. This is convenient as it allows the motion to be studied using less complicated methods. The Logistic Map helps in better understanding chaos, and gives another way to look at this phenomenon that demonstrates the diversity of to concept of chaos. The Logistic Map is important in modeling more realistic population growth, as compared to a purely exponential population growth model, which is unrealistic, as no population will have continuous exponential population growth. This model takes into account aspects that slow population growth, such as food shortage or overcrowding, and accounts for carrying capacity of the system. While these maps are not necessarily mechanical systems, studying these systems allows us to gain more understanding about mechanical systems that exhibit the same phenomena (i.e. chaos, sensitivity to initial conditions, period doubling).

Core Concepts :

LogisticMap - a function/map that gives a number x_t at regular discrete intervals as $x_t{+}1 = rx_1(1 - x_1)$

Attractor - long term response of the system; value that output will eventually converge at

Chaos - when long term motion of a system becomes non periodic

Period Doubling - the period continuously doubles and approaches infinity as γ approaches γ_c

Bifurcation Diagram - a plot of position at discrete time intervals as a function of the control parameters (in the case of the Logistic Map, this would be the growth parameter r)

Key Equations :

Logistic Map : $x_t{+}1 = rx_1(1 - x_1)$ - where $x = n/N$ is the relative population, and $(1 - x_1)$ is the "mortality factor". This equation comes from the standard growth model ($n_t{+}1 = rn_t$), and has the added mortality factor to account for the slowing of population growth.

Double Map : $g(x) = f(f(x))$, the double map take the population x_t to the expected population two years later $(x_t{+}2 = g(x_t))$

Feigenbaum Relation : $\gamma_n{+}1 - \gamma_n \approx \frac{1}{\delta}(\gamma_n - \gamma_n{-}1)$, this equation comes from the approximation for finding the distance between the threshold values/bifurcation points where period doubling occurs

Fixed Point Equation : $f(x) = x; x* = rx*(1 - x*)$, this equation comes from the fact that for a value of x to be a fixed point, we must get that value back when we plug it in to our equation. Therefore, to be a fixed point, $f(x)$ must equal x

Tricky Concepts : One concept that I struggled with was using the Feigenbaum relation to complete the derivation in part d. It took a while to get my head around how to look at the equation in a way that it could be simplified to achieve the desired outcome for the derivation. Period doubling, and the shift form period two to period four and so on, was confusing when I was first working through this section of the book. However, after working through this problem the concept became much clearer.

2 The Problem

In this case, you'll look at a slightly modified logistic map given by:

$x_(n + 1) = rx(1 - \frac{x}{2})$

(a) Look at the situation where $r = 1.5$ and your initial $x = x_0 = 0.1$:

i. Plot the value of x over the first 30 iterations. Describe it's behavior.

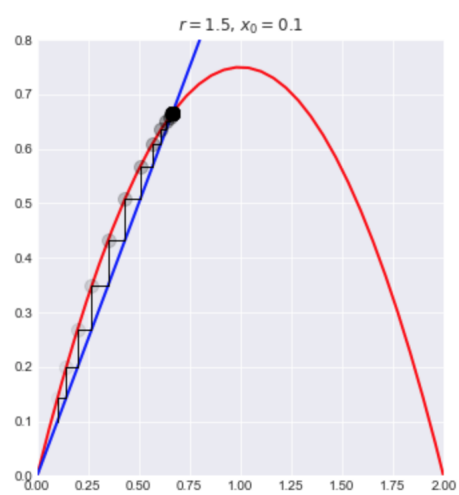

Figure 1: Plot of the Map when r=1.5

The plot of the logistic map when $r = 1.5$ shows that the system converges to the intersection point of the two plotted curves. The point where the two intersect is a fixed point. To make this plot, I created two functions: one to

define the equation for the Logistic Map, and the second to plot the system given initial values of r and x.

ii. Algebraically solve for all fixed points of the logistic map and evaluate their stability. Do your values agree with your plot?

Using the **Fixed Point Equation**, and applying it to the modified Logistic Map equation:

$$rx * (1 - \frac{x*}{2}) = x*$$

Dividing both sides by $rx*$ we get:

$$1 - \frac{x*}{2} = \frac{1}{r}$$

Subtracting $\frac{1}{r}$ from both sides, and adding $\frac{x*}{2}$, we get:

$$1 - \frac{1}{r} = \frac{x*}{2}$$

Multiplying both sides by 2

$$x = 2 - \frac{2}{r}$$

Therefore, the system has fixed points when $x = 0$ and when $x = 2 - \frac{2}{r}$

Plugging in $r = 1.5$:

$$x = 2 - \frac{2}{1.5}$$
$$x = 0.6667$$

So, in the case where $r = 1.5$, there are fixed points at $x = 0$ and $x = 0.6667$. This matches the point that is shown on the plot in Figure 1.

A fixed point is stable $|f'(x*)| < 1$, and is unstable if $|f'(x*)| > 1$ $x'_(n+1) = r - rx$, so when $x = 0$, $x' = 1.5 - 1.5(0) = 1.5$ and is **unstable** and acts as a **repeller**. When $x = 0.6667$, $x' = 1.5 - 1.5(0.6667) = 0.49995$, which is less than 1, so this fixed point is **stable** and acts as an **attractor**.

(b) Now look at the situation where $r = 3.2$. Again, you start at $x_0 = 0.1$. *i*. Plot the value of x over the first 30 iterations. Describe it's behavior. How many values does your final behavior alternate between?

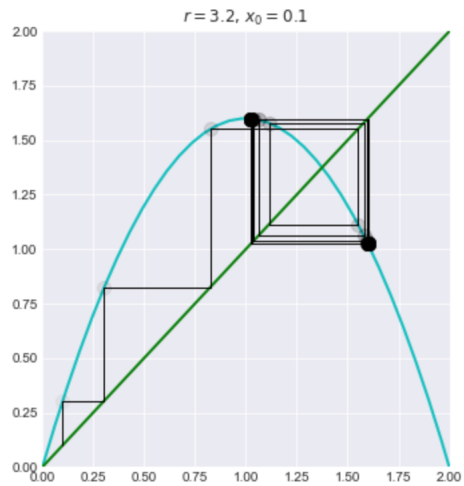

Figure 2: Plot of the Map when r=3.2

With $r = 3.2$, the system begins to exhibit chaotic behavior. Which from our definition we know to be when long term motion of the system becomes non periodic. This model, run with $r = 3.2$, has final behavior that alternates between two values, and shows the beginning of chaotic behavior. Note: these two points are fixed points of the **double map** $g(x) = f(f(x))$, and show the phenomenon of period doubling. This plot shows the model is in period two.

ii. On the different plot, plot the value of x over the first 30 interactions if $r = 3.5$ and describe the behavior. How many values does x alternate between now?

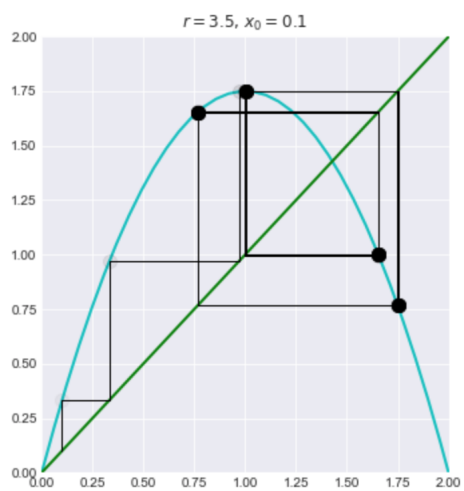

Figure 3: Plot of the Map when r=3.5

The plot of the system with $r = 3.5$ also shows chaotic behavior, as it now oscillates between four points in its end behavior. When $r = 3.5$, we see another

case of period doubling. This plot shows at $r = 3.5$, the Map is in period 4.

(c) Create a bifurcation diagram of the logistic map from $r = 2.8$ to $r = 4$. Identify several regions where the system behaves chaotically and does not settle down to a late term constant behavior.

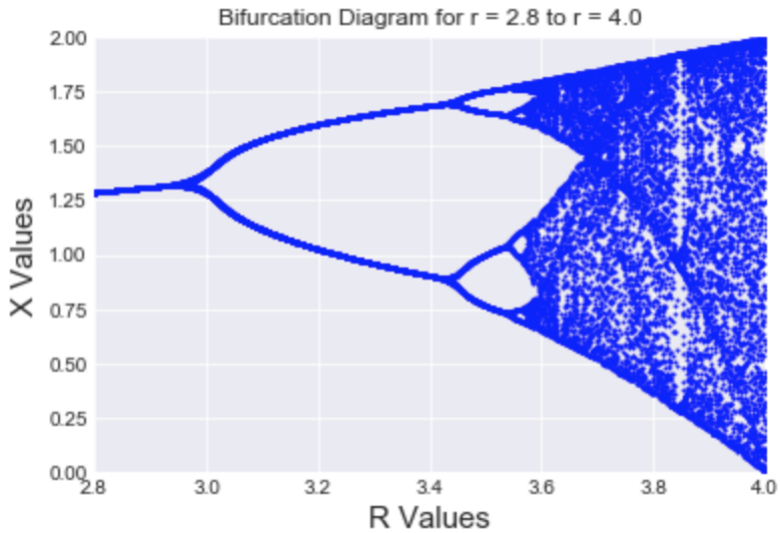

Figure 4: Bifurcation Diagram for r = 2.8 to r = 4.0

Based on Figure 4, Below are regions where system behaves chaotically:

$$2.9 \leq r \leq 3.0$$
$$3.4 \leq r \leq 3.5$$
$$3.5 \leq r \leq 3.6$$
$$r \geq 3.6$$

To find these regions, I graphed various smaller sections in python to see where the bifurcation occurs.

(d) Looking at your bifurcation diagram, you can see that the period doubling starts happening at a faster and faster rate. You can investigate this using the Feigenbaum relation (12.17 in your text).

i. Replacing the γ in 12.17 with r, show that 12.17 implies that r_n+$_1$ - $r_n = \frac{K}{\delta_n}$ where K is a constant.

Equation 12.17: **Feigenbaum Relation** : $\gamma_{n+1} - \gamma_n \approx \frac{1}{\delta}(\gamma_n - \gamma_{n-1})$

Replacing γ with r, and using the fact that r_n+$_1$ - r_n is essentially equivalent to $\frac{K}{\delta_n}$, just shifted over one value of r:

$$(r_3 - r_2) = \frac{1}{\delta}(r_2 - r_1)$$
$$(r_2 - r_1) = \frac{1}{\delta}(r_1 - r_0)$$
$$(r_1 - r_0) = \frac{1}{\delta}(r_0)$$

Since r_0 is a constant value, this shows: r_n+$_1$ - $r_n = \frac{K}{\delta_n}$, where K is the constant value r_0.

Therefore, γ_n+1 $- \gamma_n \approx \frac{1}{\delta}(\gamma_n - \gamma_n$−1) implies r_n+$_1$ - $r_n = \frac{K}{\delta_n}$.

ii. Using your bifurcation diagram as a guide, pinpoint to 3 decimals the points when the period doubles from 1 to 2, 2 to 4, and 4 to 8. Make a plot of $ln(r_n$+$_1$ - $r_n)$ against n. The best-fit straight line to the data has a slope equivalent to the Feigenbaum constant δ. Find this value. This value describes how fast the period doubling cascade happens as r varies.

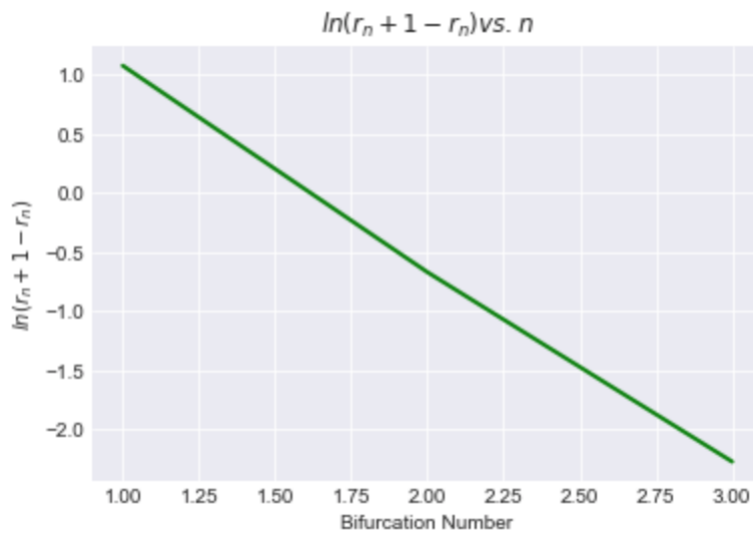

Figure 5: Graph of $ln(r_n$+$_1$ - $r_n)$ against n

Period doubles from 1 to 2 at: $r \approx 2.929$
Period doubles from 2 to 4 at: $r \approx 3.440$
Period doubles from 4 to 8 at: $r \approx 3.543$
I found these values by plotting the bifurcation diagram for very small intervals of r and x.

I used the values listed above, as well as the number (1-3) of each bifurcation, to generate the plot shown in Figure 5. The slope of this plot is -1.746, which means the Feigenbaum constant for this system is $\delta = 1.746$. This is the speed with which period doubling occurs with varied r.

References

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